



Aishu R 251001007-2025-IT ▾

A2**Started on** Tuesday, 4 August 2026, 8:59 AM**State** Finished**Completed on** Tuesday, 4 August 2026, 9:00 AM**Time taken** 54 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num;++i)
    {
        if (num % i== 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 | #include<stdio.h>
2 | void Factor( int num) {
3 |     int count=0;
4 |     //count=count+1;
5 |     for (int i = 1; i <= num;++i)
6 |     {
7 |         count=count+2;
8 |         if (num % i== 0)
9 |         {
10 |             count=count+1;
11 |             //printf("%d ", i);
12 |         }
13 |     }
14 |     count=count+1;
15 |     printf("%d",count);
16 | }
17 | int main(){
18 |     int n;
19 |     scanf("%d",&n);
20 |     Factor(n);
21 |     return 0;
22 | }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! 

Correct

Marks for this submission: 1.00/1.00.

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